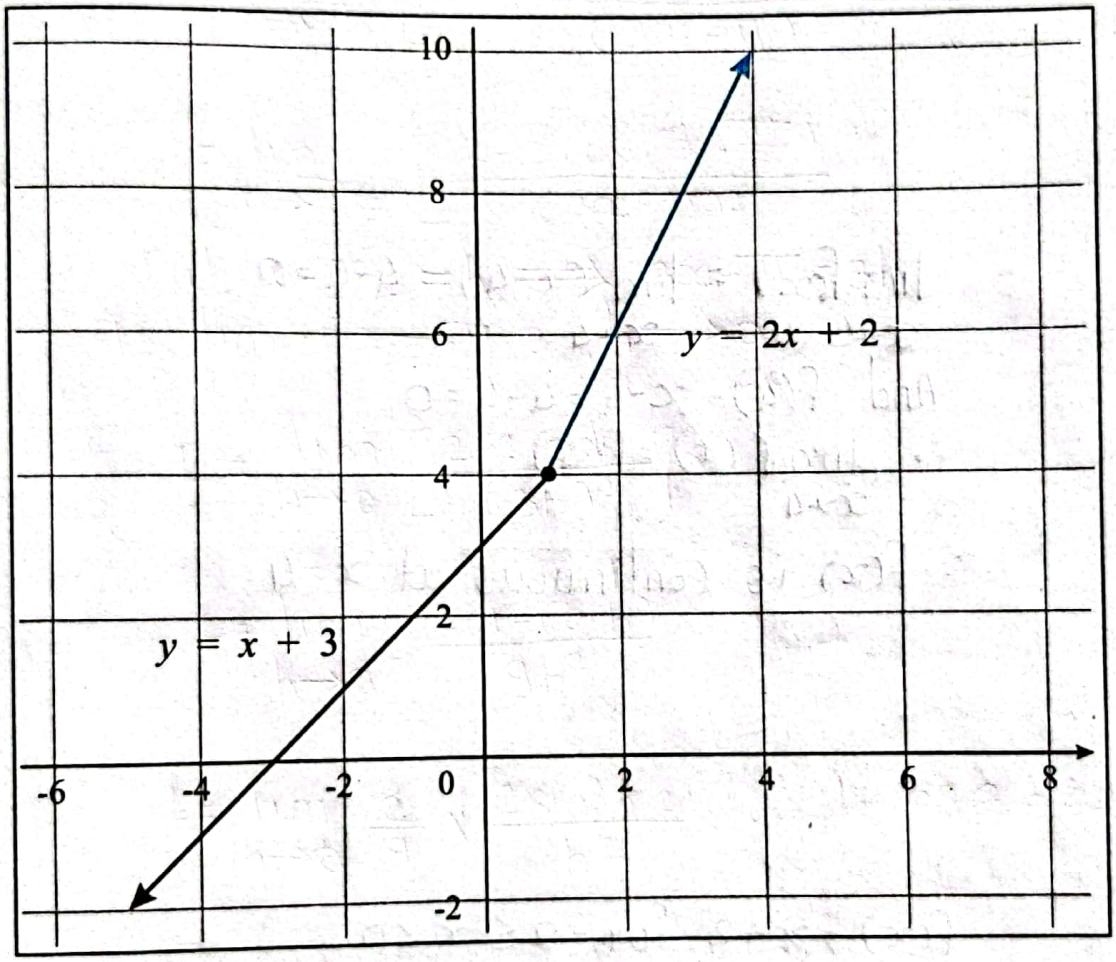
(1) By drawing the graph of $f(x)$, test whether the function $f(x)=\begin{cases} x+3, & for\ x<1 \\ 2x+2, & for\ x\geq 1 \end{cases}$ is continuous at $x=1$. Justify your answer. Identify the type of discontinuity if it is discontinuous.

Solution : Take points on the line $y=x+3$. Plot these points. Draw part of the line for $x<1$. Take points on the line $y=2x+2$. Plot these points. Draw part of the line for $x\geq 1$. The graph is continuous. There is no break, gap in this graph. From the graph we observe that $f(x)$ is continuous at $x=1$



2) Test the continuity of $f(x)=\begin{cases} x^2-3, & for\, 0\leq x\leq 2 \\ x-4, & for\, 2<x\leq 4 \end{cases}$ at a) $x = 1$ b) $x = 2$ c) $x = 4$

Solution : a) To discuss the continuity of $f(x)$ at $x=1$.We observe that the definition of $f(x)$ is uniform (same) in the neighbourhood of 1.

$\lim\limits_{x\to 1} f(x)=\lim\limits_{x\to 1}(x^2-3)=-2$

And $f(1)=(1)^2-3=-2$

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$\therefore \lim_{x \to 1} f(x) = f(1)$

$\therefore f(x)$ is continuous at $x = 1$.

b) To discuss the continuity of $f(x)$ at $x = 2$ We observe that the definition of $f(x)$ is not uniform (same) in the neighbourhood of 2. We have to evaluate the left and right limits separately.

$\lim_{x \to 2^+} f(x) = \lim_{x \to 2} (x - 4) = -2$ $\therefore$ The right hand side limit is -2.

$\lim_{x \to 2^-} f(x) = \lim_{x \to 2} (x^2 - 3) = 1$ $\therefore$ The left hand side limit is 1.

$\lim_{x \to 2^+} f(x) \neq \lim_{x \to 2^-} f(x)$ i.e. the right hand side limit is not equal to the left hand side limit.

$\therefore$ Function $f(x)$ is not continuous at $x = 2$.

c) To discuss the continuity of $f(x)$ at $x = 4$. We observe that the definition of $f(x)$ is uniform (same) in the neighbourhood of 4.

$\lim_{x \to 4} f(x) = \lim_{x \to 4} (x-4) = 4-4 = 0$

And $f(4) = x - 4 = 4 - 4 = 0$

$\therefore \lim_{x \to 4} f(x) = f(4)$

$\therefore f(x)$ is continuous at $x = 4$

3) Draw the graph of $f(x) = \begin{cases} x+2, & for -2 \le x < 0 \\ 0 & for x = 0 \\ -x+2, & for 0 < x \le 2 \end{cases}$

Is f continuous at $= 0$?

Solution : $f(x) = x + 2$, for $-2 \le x < 0$

when $x = -2$, $f(-2) = -2 + 2 = 0$ point is $(-2, 0)$

when $x = -1$, $f(-1) = -1 + 2 = 1$ point is $(-1, 1)$

$f(x) = 0$, for $x = 0$ point is $(0, 0)$

$f(x) = -x + 2$, for $0 < x \le 2$

when $x = 1$, $f(1) = -1 + 2 = 1$ point is $(1, 1)$

when $x = 2$, $f(2) = -2 + 2 = 0$ Point is $(2, 0)$

From graph $f(x)$ is discontinuous at $x = 0$ because graph is distinct at $(0, 0)$

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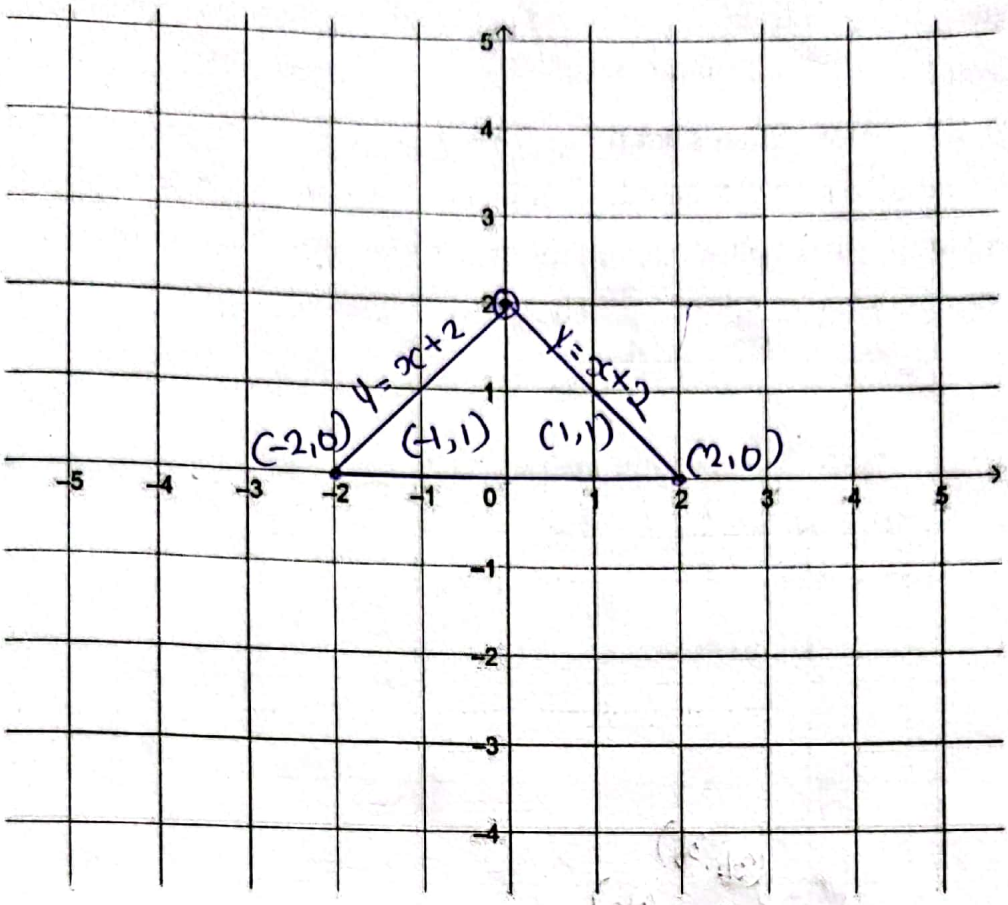

4) Test the continuity of $f(x)=\begin{cases} x^2, & for -1 \le x \le 0 \\ x & for 0 < x \le 3 \end{cases}$ at $x=0$.

Solution : To discuss the continuity of $f(x)$ at $x=0$. We observe that the definition of $f(x)$ is not uniform (same) in the neighbourhood of 0. We have to evaluate the left and right limits separately.

$\lim_{x \to 0^-} f(x) = \lim_{x \to 0^-} (x^2) = (0)^2 = 0$

$\lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} (x) = 0$

The left hand limit and the right hand limit are equal

$\lim_{x \to 0^-} f(x) = \lim_{x \to 0^+} f(x)$

Also $f(0) = 0^2 = 0$

$\therefore$ The given function $f(x)$ is continuous at $x=0$

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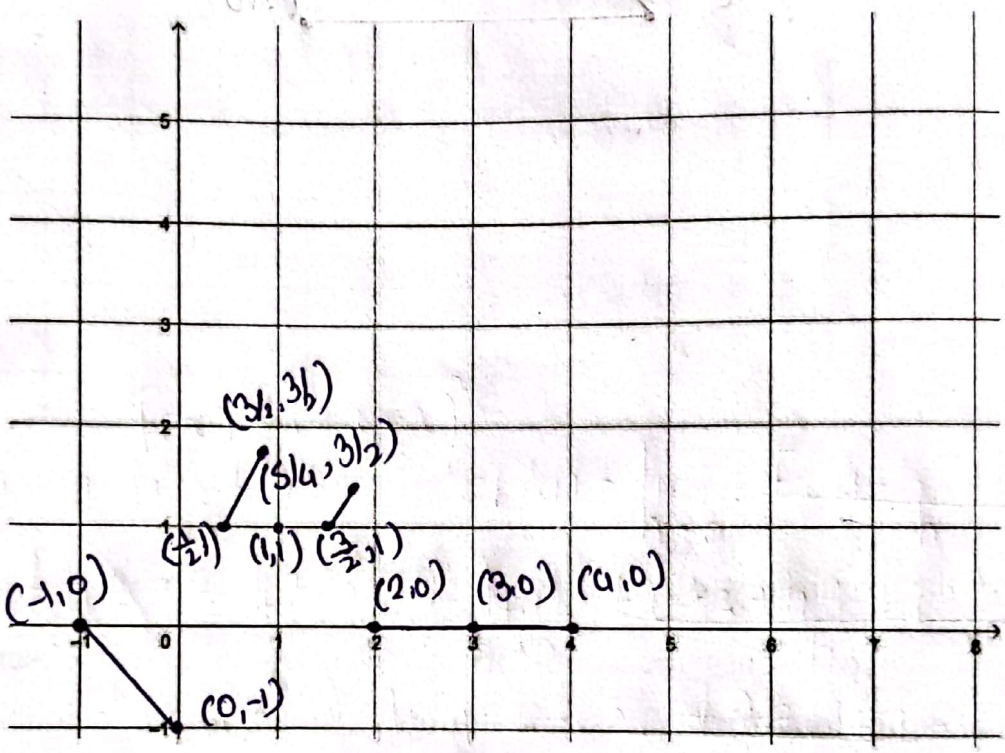
5) Draw the graph of function $f(x)$ and discuss its continuity on its domain.

$$f(x)=\begin{cases} x^2-1, & if\ -1\le x\le 0 \\ 2x, & if\ 0<x<1 \\ 1, & if\ x=1 \\ 4-2x, & if\ 1<x<2 \\ 0, & if\ 2\le x\le 4 \end{cases}$$

Solution :



$f(x)=x^2-1,\ -1\le x\le 0$

for $x=-1$, $f(-1)=1-1=0$, for $x=0$, $f(0)=0-1=-1$

Point is $(-1,0)$ Point is $(0,-1)$

$f(x)=2x,\ 0<x<1$

For $x=1/2$, $f(1/2)=2\times 1/2=1$, for $x=3/4$, $f(3/4)=2\times 3/4=3/2$

Point is $(1/2,1)$ Point is $(3/4,3/2)$

$f(x)=1$, if $x=1$, Point is $(1,1)$

$f(x)=4-2x,\ 1<x<2$

for $x=3/2$, $f(3/2)=4-2\times 3/2=1$ for $x=5/4$, $f(5/4)=3/2$

Point is $(3/2,1)$ Point is $(5/4,3/2)$

$f(x)=0$ for $2\le x\le 4$ Point are $(2,0)$, $(3,0)$, $(4,0)$

The functions is not continuous at $x=0$, $x=1$, $x=2$

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6) If $f(x)=\begin{cases} x^2-1, & for -3 \le x < 0 \\ px+q, & for 0 \le x \le 1 \\ x+1, & for 1 < x \le 3 \end{cases}$ is continuous on its domain then find values of p and q.

Solution : Since f(x) is Continuous on its domain

f(x) is Continuous at $x=0$ and at $x=1$

Let f(x) is Continuous at $x=0$

$\therefore \lim_{x \to 0} f(x) = \lim_{x \to 0^+} (Px+q)$

$\therefore (0-1) = (P \times 0 + q)$

$\therefore -1 = q$

$\boxed{q=-1}$

Let f(x) is Continuous at $x=1$

$\therefore \lim_{x \to 1^-} f(x) = \lim_{x \to 1^+} f(x)$

$\therefore P+q = 1+1$

$\therefore P+q = 2$

$\therefore P-1 = 2 \quad (\because q=-1)$

$\therefore \boxed{P=3}$

$\therefore P=3, \; q=-1$

Sign of Teacher :

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